

Singular-type operators preserve the Morrey vanishing property (V^*)

Candidate full answer to the question in arXiv:1911.06551

17 August 2026

Status: candidate full proof, likely valid. For every $1 < p < \infty$ and $0 \leq \lambda < n$, every p -admissible sublinear singular-type operator preserves $V^{(*)}L^{p,\lambda}(\mathbb{R}^n)$. This includes all singular integral operators asked about in the source.

1 Source question and definitions

For $1 < p < \infty$ and $0 \leq \lambda < n$, the Morrey space $L^{p,\lambda}(\mathbb{R}^n)$ has norm

$$\|f\|_{p,\lambda} := \sup_{x \in \mathbb{R}^n, r > 0} \left(r^{-\lambda} \int_{B(x,r)} |f(y)|^p dy \right)^{1/p}.$$

Writing $\chi_N = \mathbf{1}_{\mathbb{R}^n \setminus B(0,N)}$, define

$$\mathcal{A}_{N,p}(f) := \sup_{x \in \mathbb{R}^n} \int_{B(x,1)} |f(y)|^p \chi_N(y) dy.$$

The subspace $V^{(*)}L^{p,\lambda}(\mathbb{R}^n)$ consists of the $f \in L^{p,\lambda}(\mathbb{R}^n)$ satisfying $\mathcal{A}_{N,p}(f) \rightarrow 0$.

The source calls a sublinear operator T *p-admissible singular-type* if it is bounded on $L^p(\mathbb{R}^n)$ and, for $z \in B(x,r)$,

$$|T(f\mathbf{1}_{\mathbb{R}^n \setminus B(x,2r)})(z)| \leq C_T \int_{\mathbb{R}^n \setminus B(x,2r)} \frac{|f(y)|}{|y-z|^n} dy. \quad (1)$$

Its Lemma 2.3 implies that such T is bounded on $L^{p,\lambda}(\mathbb{R}^n)$. Immediately before Theorem 3.11, the authors ask whether a singular integral S in this class preserves (V^*) [1, p. 9].

We end this section with a further result for singular integral operators S defined by (2.3), with the kernel satisfying (2.4). One knows that S is bounded in $V_0L^{p,\lambda}(\mathbb{R}^n)$, for $1 < p < \infty$ and $0 \leq \lambda < n$ (cf. [32] Theorem 5.1). On the other hand, we have seen that S is also bounded in $V_\infty L^{p,\lambda}(\mathbb{R}^n)$ (cf. Corollary 3.2). Unfortunately, we do not know whether S preserves the vanishing property (V^*) . Nevertheless, we have the following result:

Theorem 3.11. *Let $1 < p < \infty$ and $0 \leq \lambda < n$. Then S is bounded from $V_{0,\infty}^{(*)}L^{p,\lambda}(\mathbb{R}^n)$ into itself.*

Figure 1: The exact open question on page 9 of the arXiv PDF.

2 Proof intuition

The size condition reduces the only difficult piece to an order- n tail potential. On a dyadic annulus of radius 2^k , the (V^*) hypothesis gives a bound by the small far-unit-ball quantity η_N , but this bound alone is not summable in k . The Morrey norm gives the complementary bound $M2^{-k(n-\lambda)/p}$. Taking the better estimate on every annulus and summing gives

$$\sum_{k \geq 0} \min\{\eta_N, M2^{-k(n-\lambda)/p}\} \lesssim \eta_N (1 + \log(M/\eta_N)) \longrightarrow 0.$$

This logarithmic interpolation is the missing critical-tail mechanism.

3 Full positive answer

Lemma 1 (Critical tail estimate). *Let $f \in L^{p,\lambda}(\mathbb{R}^n)$, $M = \|f\|_{p,\lambda}$, and*

$$F_N = f \mathbf{1}_{\mathbb{R}^n \setminus B(0,N)}, \quad \eta_N = \sup_{u \in \mathbb{R}^n} \|F_N\|_{L^p(B(u,1))}.$$

Set $a = (n - \lambda)/p > 0$. If G is any measurable function with $|G| \leq |F_N|$ and $G = 0$ on $B(x, 2)$, then, uniformly for $z \in B(x, 1)$,

$$\int_{\mathbb{R}^n} \frac{|G(y)|}{|y - z|^n} dy \leq C_{n,p,\lambda} \eta_N \left(1 + \log_+ \frac{M}{\eta_N}\right), \quad (2)$$

with the right side interpreted as 0 when $\eta_N = 0$.

Proof. Every ball of radius $R \geq 1$ can be covered by at most $C_n R^n$ unit balls. Since multiplication by an indicator does not increase the Morrey norm,

$$\|G\|_{L^p(B(z,R))} \leq C_n \min\{\eta_N R^{n/p}, MR^{\lambda/p}\}. \quad (3)$$

The support condition and $z \in B(x, 1)$ imply $|y - z| \geq 1$ on the support of G . Split that support into the annuli $2^k \leq |y - z| < 2^{k+1}$, $k \geq 0$. Hölder's inequality and (3) yield

$$\int_{2^k \leq |y-z| < 2^{k+1}} \frac{|G(y)|}{|y - z|^n} dy \leq C_{n,p} \min\{\eta_N, M2^{-ak}\}.$$

Here the first alternative follows because the volume factor $2^{kn/p'}$ and the unit-cover factor $2^{kn/p}$ exactly cancel the kernel 2^{-kn} ; the second leaves $2^{-k(n-\lambda)/p}$.

If $0 < \eta_N \leq M$, let

$$K = \left\lceil \frac{\log(M/\eta_N)}{a \log 2} \right\rceil.$$

The terms with $k < K$ total at most $K\eta_N$, while the remaining geometric tail is at most $M2^{-aK}/(1 - 2^{-a}) \leq \eta_N/(1 - 2^{-a})$. This proves (2). The cases $M = 0$ or $\eta_N = 0$ are immediate. (In fact $\eta_N \leq M$ by the Morrey estimate at radius one.) $\square$

Theorem 2. *Let $1 < p < \infty$ and $0 \leq \lambda < n$. Every p -admissible sublinear singular-type operator T maps $V^{(*)}L^{p,\lambda}(\mathbb{R}^n)$ boundedly into itself. In particular, every singular integral operator S considered in the source preserves the vanishing property (V^*) .*

Proof. The source's Morrey boundedness result already gives $Tf \in L^{p,\lambda}(\mathbb{R}^n)$ and $\|Tf\|_{p,\lambda} \lesssim \|f\|_{p,\lambda}$. It remains to prove (V^*) .

Fix $f \in V^{(*)}L^{p,\lambda}(\mathbb{R}^n)$, write $M = \|f\|_{p,\lambda}$, and use η_N from Lemma 1. Thus $\eta_N = \mathcal{A}_{N,p}(f)^{1/p} \rightarrow 0$. For fixed N and $|x| > N + 3$, decompose

$$f = u_x + v_N + w_{N,x},$$

where

$$u_x = f\mathbf{1}_{B(x,2)}, \quad v_N = f\mathbf{1}_{B(0,N)}, \quad w_{N,x} = f\mathbf{1}_{(\mathbb{R}^n \setminus B(0,N)) \setminus B(x,2)}.$$

The first two supporting balls are disjoint, so these three pieces cover f . By sublinearity and Minkowski's inequality it suffices to estimate the three outputs on $B(x,1)$.

First, the global L^p boundedness of T and a fixed finite cover of $B(x,2)$ by unit balls give

$$\|Tu_x\|_{L^p(B(x,1))} \leq C_T \|u_x\|_{L^p(\mathbb{R}^n)} \leq C_{n,T}\eta_N, \quad (4)$$

because $B(x,2) \subset \mathbb{R}^n \setminus B(0,N)$.

Second, v_N is supported off $B(x,2)$. For $z \in B(x,1)$, the size condition gives

$$|Tv_N(z)| \leq C_T \int_{B(0,N)} \frac{|f(y)|}{|y-z|^n} dy \leq \frac{C_T \|f\mathbf{1}_{B(0,N)}\|_{L^1}}{(|x| - N - 1)^n}.$$

Consequently, for each fixed N ,

$$\|Tv_N\|_{L^p(B(x,1))} \rightarrow 0 \quad (|x| \rightarrow \infty). \quad (5)$$

Finally, $w_{N,x}$ satisfies the hypotheses of Lemma 1; applying the size condition and then the lemma gives

$$\|Tw_{N,x}\|_{L^p(B(x,1))} \leq C_{n,p,\lambda,T}\eta_N \left(1 + \log_+ \frac{M}{\eta_N}\right). \quad (6)$$

From (4)–(6),

$$\limsup_{|x| \rightarrow \infty} \|Tf\|_{L^p(B(x,1))} \leq C\eta_N \left(1 + \log_+ \frac{M}{\eta_N}\right).$$

Letting $N \rightarrow \infty$ makes the right side zero. Hence the L^p norm of Tf on unit balls tends to zero uniformly as their centers go to infinity.

To recover the source's cut-off formulation, note that $B(x,1) \cap (\mathbb{R}^n \setminus B(0,R)) \neq \emptyset$ implies $|x| > R - 1$. Therefore

$$\mathcal{A}_{R,p}(Tf)^{1/p} \leq \sup_{|x| > R-1} \|Tf\|_{L^p(B(x,1))} \rightarrow 0.$$

Thus $Tf \in V^{(*)}L^{p,\lambda}(\mathbb{R}^n)$. □

4 Scope, verification, and novelty

The theorem uses exactly the source definition of a p -admissible singular-type operator. It does not require linearity, kernel smoothness, or additional cancellation. The strict inequality $\lambda < n$ is essential to this argument because it makes $a = (n - \lambda)/p$ positive. All source parameters, including $\lambda = 0$, are covered.

The dyadic transition and vanishing modulus were stress-tested over 72,000 parameter triples by the included verifier; this is a check of the algebra, not a proof. The formal proof above supplies the exact geometric-series argument.

A bounded novelty search on 17 August 2026 covered the local run indexes, the arXiv identifier and title, the exact phrase “we do not know whether S preserves the vanishing property (V^*) ”, the authors, singular operators on vanishing Morrey spaces, and later citing papers. The closest item is the same authors’ generalized-Morrey follow-up [2]. Crucially, its Section 5 again states the same uncertainty and its Theorem 5.3 only proves invariance of the closure of C_0^∞ , not invariance of $V^{(*)}$ itself. Searches found no later full answer or matching logarithmic dyadic-tail argument. Novelty confidence is moderate rather than high because the argument is elementary and the search was bounded.

Human-review recommendation. Check (i) that the source size condition applies to the noncompact truncated tail $w_{N,x}$ under its standard Morrey extension, (ii) the annular Hölder exponents in Lemma 1, and (iii) the last equivalence between far-center decay and the cut-off definition of (V^*) .

References

- [1] A. Ç. Albalik, A. Almeida, and S. Samko, *On the invariance of certain vanishing subspaces of Morrey spaces with respect to some classical operators*, Banach J. Math. Anal. 14 (2020), 987–1000; arXiv:1911.06551, <https://arxiv.org/abs/1911.06551>.
- [2] A. Ç. Albalik, A. Almeida, and S. Samko, *Preservation of certain vanishing properties of generalized Morrey spaces by some classical operators*, Math. Methods Appl. Sci. 43 (2020), 9375–9386, doi:10.1002/mma.6235.